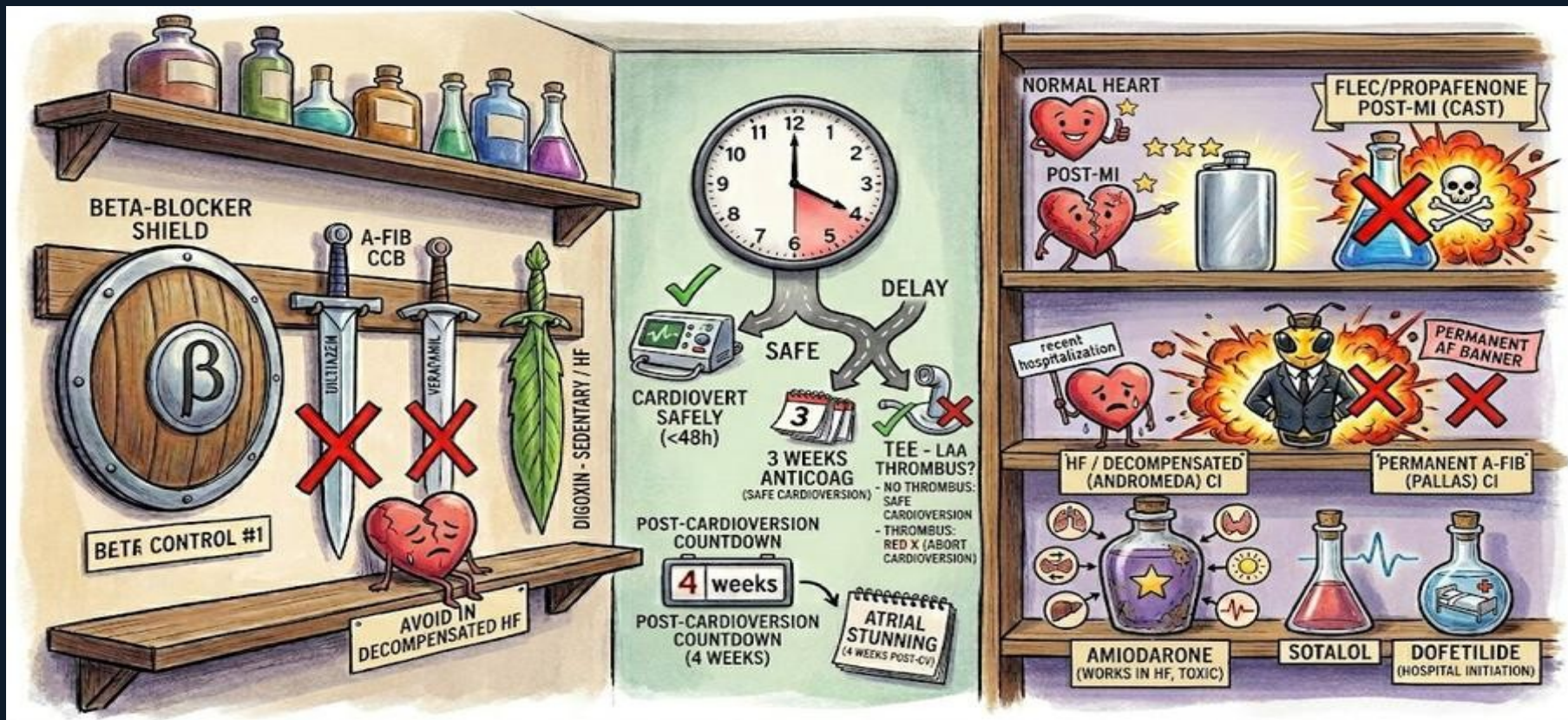


Scene D: AF Rate Control, Rhythm Control & Cardioversion



Armory & Apothecary • MUST MEMORIZE • 6 Board Questions

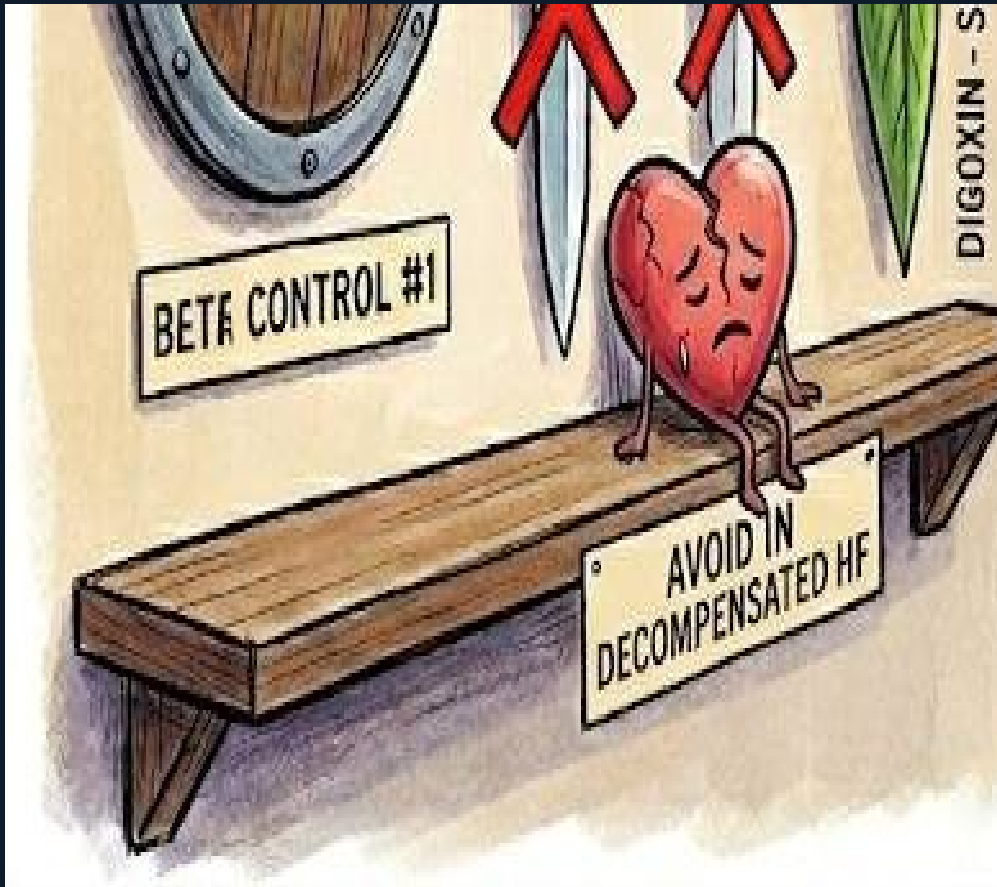


Beta-Blocker Shield — Rate Control #1

The beta-blocker shield (β symbol) hangs prominently as the primary rate control weapon, labeled 'RATE CONTROL #1.'

Beta-blockers (metoprolol, atenolol, carvedilol) are first-line for rate control in most AF patients. They are safe in stable heart failure (carvedilol, metoprolol succinate), hypertension, and CAD. Target resting HR <110 (lenient) or <80 (strict/symptomatic).

Board trap: choosing diltiazem as first-line rate control in a patient with decompensated HFrEF. Beta-blockers tolerated (with caution); diltiazem/verapamil avoided.



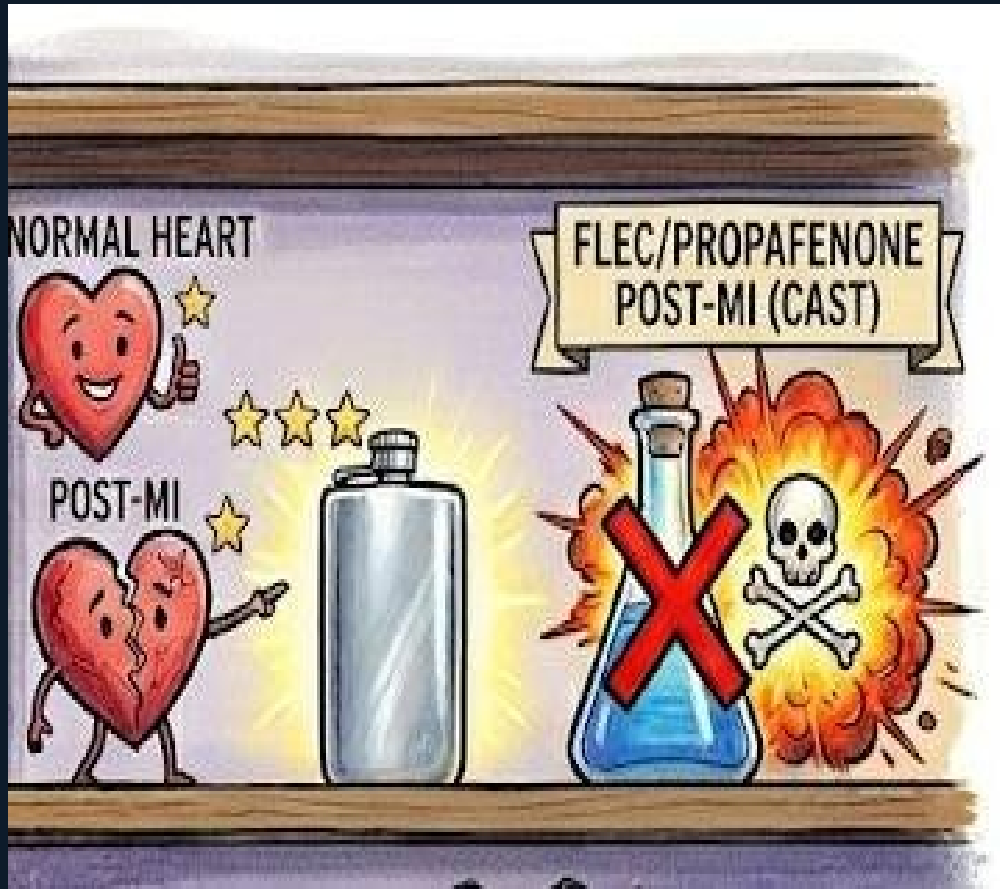
Scene D — AF Rate/Rhythm Control + Cardioversion | Arrhythmias Board Prep

CCBs — Avoid in Decompensated HF

Diltiazem and verapamil swords hang on the rack. A cracked heart slumps beneath them. Both swords have red Xs when aimed at this heart. Label: 'AVOID IN DECOMPENSATED HFrEF.'

Nondihydropyridine CCBs (diltiazem, verapamil) are negative inotropes — they reduce cardiac contractility. In decompensated HFrEF (low EF, fluid overload), they worsen cardiac output and can precipitate acute decompensation.

Board trap: 'use diltiazem for rate control in AF with EF 25%' — wrong. Use beta-blocker, digoxin, or amiodarone in this setting.

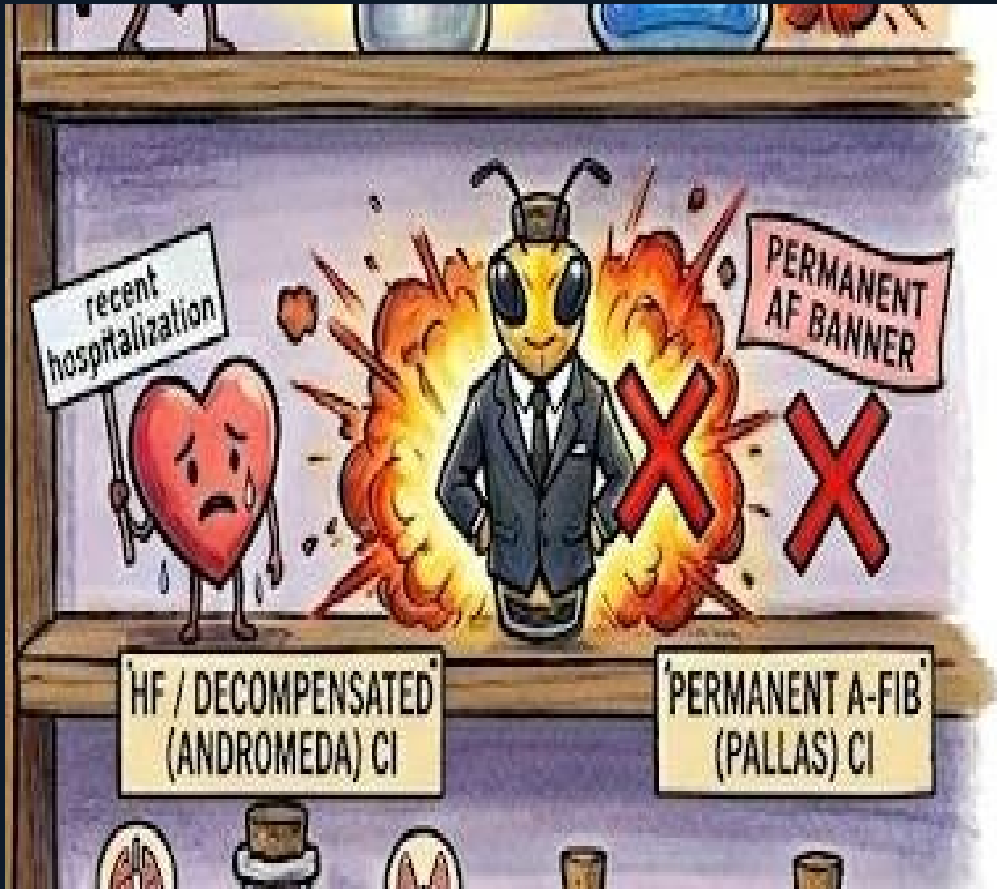


Flecainide Post-MI — Explosion

Class IC flasks (flecainide, propafenone) glow with gold stars near a normal heart — but when a post-MI scar heart approaches, an explosion erupts with skull and red X. Label: 'FLEC/PROPAFENONE POST-MI (CAST) = INCREASED MORTALITY.'

The CAST trial showed class IC drugs increased mortality in post-MI patients despite suppressing arrhythmias. Class IC drugs are proarrhythmic on ischemic myocardium.

Rule: class IC drugs for rhythm control only in structurally normal hearts. Any structural heart disease (MI, HF, significant LVH) = contraindicated.

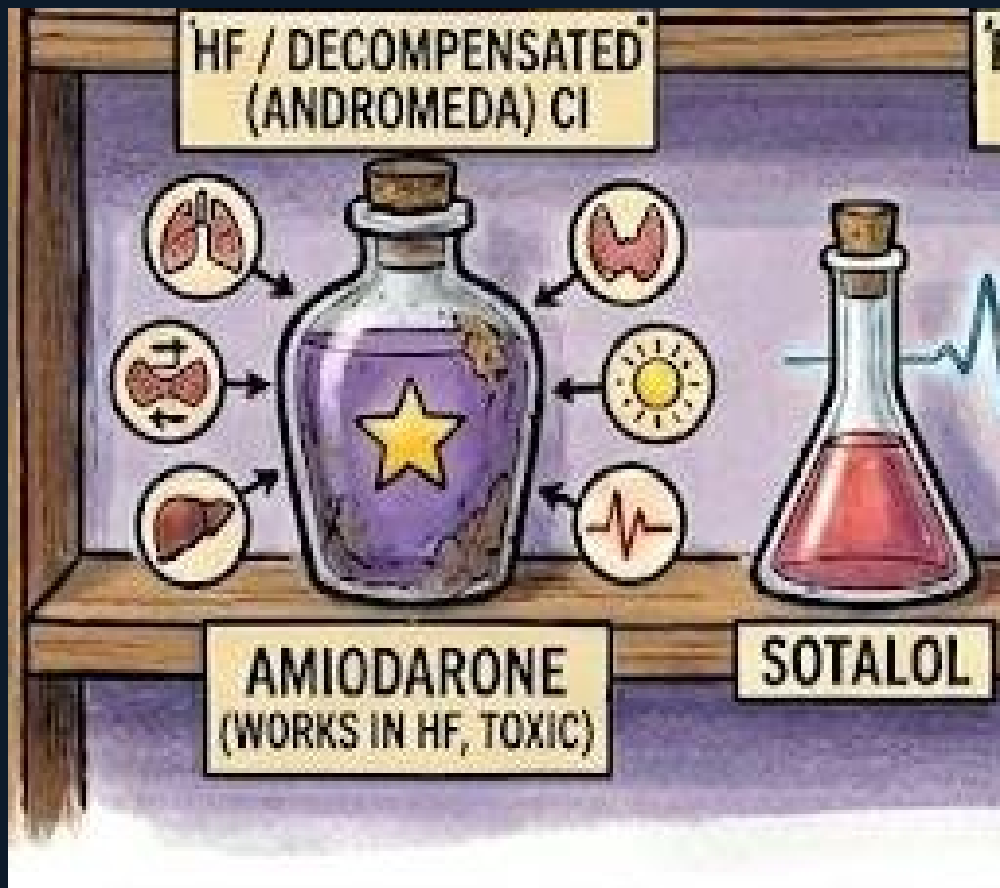


Dronedarone — Two Fatal Contraindications

The drone-bee (dronedarone) character shows two explosion scenes: one near a decompensated heart (ANDROMEDA trial) and one near a 'PERMANENT AF' banner (PALLAS trial). Both have red Xs.

Dronedarone (Multaq) increases mortality in: (1) decompensated or recently hospitalized HF patients (ANDROMEDA trial), and (2) patients with permanent AF (PALLAS trial).

Board trap: prescribing dronedarone to a patient with recent HF hospitalization or permanent AF. These are absolute contraindications. Dronedarone is only appropriate in stable paroxysmal/persistent AF with preserved LV function.



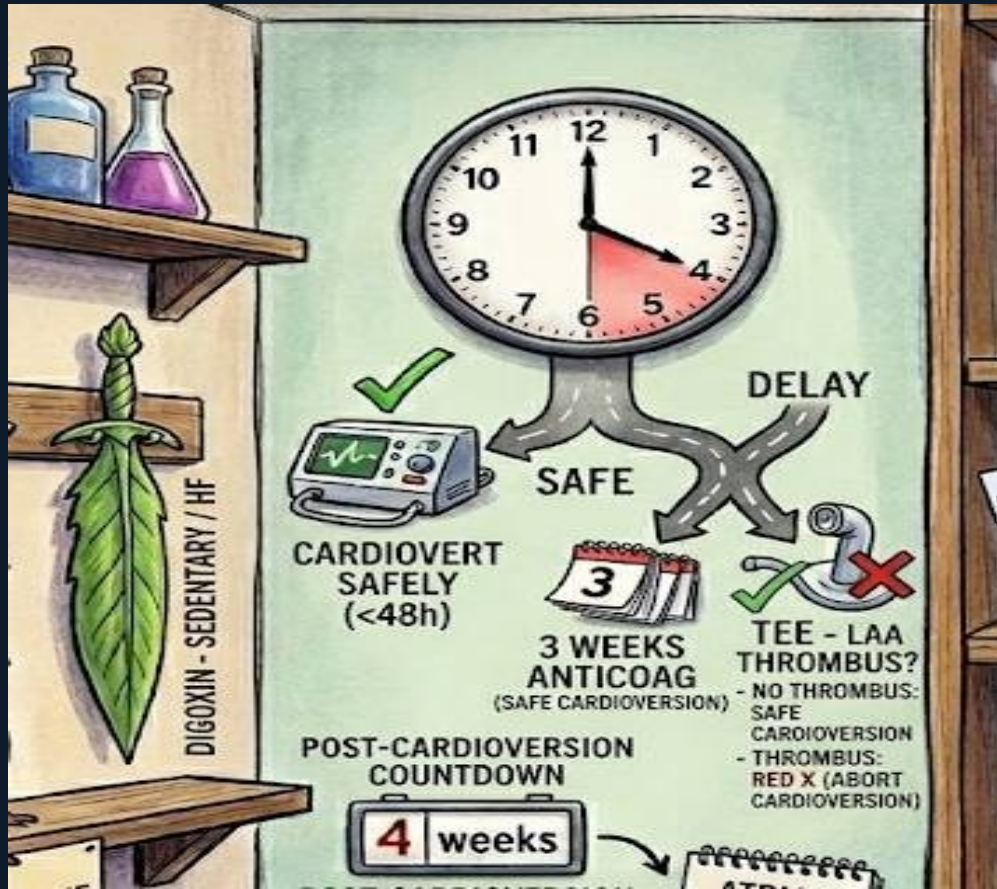
Scene D — AF Rate/Rhythm Control + Cardioversion | Arrhythmias Board Prep

Amiodarone — Works in HF, But Toxic

A large weathered purple bottle covered in toxicity icons (lung cracks = pulmonary fibrosis, thyroid arrows = thyroid dysfunction, liver spots, corneal deposits, sun rays = photosensitivity, bradycardia line) holds a gold star. Label: 'WORKS IN HF — BUT TOXIC.'

Amiodarone is the most effective antiarrhythmic for maintaining sinus rhythm and can be used in any patient including those with heart failure. But its long-term toxicity profile is extensive.

Board trap: 'amiodarone is contraindicated in HF' — wrong. It works in HF. The board tests amiodarone toxicities: pulmonary fibrosis (cough, SOB), thyroid (hypo or hyper), corneal deposits, liver toxicity, photosensitivity, bradycardia, QT prolongation.



Scene D — AF Rate/Rhythm Control + Cardioversion | Arrhythmias Board Prep

The 48-Hour Clock

A large clock shows the 48-hour dividing line. Left of 48h: cardioversion fires freely with a green checkmark. Right of 48h and 'UNKNOWN DURATION': two paths fork — 3 calendar pages (3 weeks anticoagulation) or a TEE probe (exclude LAA thrombus).

AF <48h: can cardiovert without prior anticoagulation (low thrombus risk). AF >48h or unknown duration: must either anticoagulate for 3 weeks before cardioversion, or perform TEE to rule out LAA thrombus before cardioverting.

Board trap: cardioverting AF of unknown duration without 3 weeks anticoagulation and without TEE — wrong. Risk of dislodging a formed LAA thrombus → stroke.



Scene D — AF Rate/Rhythm Control + Cardioversion | Arrhythmias Board Prep

Post-Cardioversion: 4 Weeks Mandatory

A 4-week countdown timer shows 'ANTICOAGULATE ≥ 4 WEEKS POST-CARDIOVERSION.' Label: 'ATRIAL STUNNING.'

After cardioversion, the atrial myocardium is mechanically stunned and cannot contract effectively for up to 4 weeks — even though the ECG shows sinus rhythm. During this period, thrombus can still form and embolize.

Board trap: 'patient cardioverted from AF <48h, no anticoagulation needed afterward' — wrong. Post-cardioversion anticoagulation for ≥ 4 weeks is mandatory regardless of AF duration, unless CHA₂DS₂-VASc is 0 in men or 1 in women.

Scene D — Board Trap Master Summary

Beta-blocker = rate control #1

Safe in stable HF; first-line for most

Flecainide/propafenone post-MI → mortality ↑

CAST trial; only for structurally normal hearts

Dronedarone: CI in permanent AF (PALLAS)

Increases mortality; only paroxysmal/persistent

AF >48h or unknown: 3wk anticoag OR TEE

Before cardioversion; prevent stroke

Sotalol: check QT + renal function

Prolongs QT; avoid if CrCl <40

Diltiazem/verapamil: avoid in decomp HFrEF

Negative inotropy worsens cardiac output

Dronedarone: CI in decomp HF (ANDROMEDA)

Increases mortality in recent HF hospitalization

Amiodarone: works in HF despite toxicity

Most effective; know all 6 toxicities

Post-cardioversion: ≥4 weeks anticoag

Atrial stunning regardless of AF duration

Dofetilide: in-hospital initiation only

QT monitoring required; hospital initiation